



Assignment 2_PERMUTATION & COMBINATION

Please watch the video lesson named "Permutation Combination assignment" for the solution.

1. How many numbers of 5 digits can be formed with the digits 4, 6, 7, 5, 2.

(a) If no digit being repeated.

Ans : 120

(b) If repetition of digit is allowed.

Ans : 3125

2. How many numbers of 5 digits can be formed with the digits 2, 7, 0, 5, 4 ?

Ans : 96

3. How many numbers of 4 digits can be formed with the digits 2, 3, 5, 7, 8 and 9 ?

a) If repetition of digit is not allowed:

Ans : 360

b) If repetition of digit is allowed:

Ans: 1296

4. How many numbers of 4 digits can be formed with the digits 0, 2, 3, 4, 6, 8 and 9 ?

a) If repetition of digit is not allowed:

Ans : 720

b) If repetition of digit is allowed:

Ans : 2058

5. How many numbers greater than 7000 can be formed with four of the digits 4, 5, 7, 8, and 9?

Ans : 72

6. How many numbers greater than 500 and less than 3000 can be formed with the digits 0, 1, 2, 3, 4, 5 and 6?

Ans : 300



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7. How many such 5 digit numbers which are exactly divisible by 5 can be formed with the digits 1, 4, 5, 6, 7, 8 and 9?

Ans : 360

8. How many such 5 digit numbers which are exactly divisible by 5 can be formed with the digits 0, 2, 3, 5, 6, 7 and 8?

Ans : 660

9. In how many ways, can a

(a) team of six be chosen out of nine students?

Ans : 84

(b) A team of six to be chosen out of twelve students?

Ans : 924

10. In how ways a committee of six members can be selected from seven men and six women

(a) Consisting of three men and three women?

Ans : 700

(b) Consisting of two men and four women?

Ans : 315

11. There are 7 members in a delegation which is to be sent abroad. The total no. of members is 12. In how ways can the selection be made so that a particular member is always

(i) included

Ans : 462

(ii) excluded?

Ans : 330

12. A committee of some members is to be made out of a group of 9 men and 5 women. In how many different ways can it be made if the committee is to be made according to the following stipulation



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(a) the committee of 4 must include at least one women.

Ans : 875

(b) the committee of 8 must have exactly four men and four women .

Ans : 630

13. A committee of 6 students is to be formed out of 4 boys and 7 girls. In how many ways can it be done so that the no. of girls is not less than the no. of boys?

Ans : 441

14. In how many ways, 3 girls and 5 boys are to be seated in a row so that no girl is ever together.

Ans :1440